

VALENTIA OBSERVATORY, Ireland

WMO: 039530

Lat: 51.9383N

Lon: 10.2410W

Elev: 98

StdP: 14.64

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.4	34.1	24.2	18.4	35.4	27.0	21.0	37.6	37.0	46.0	32.8	46.9	4.6	60	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	70.1	61.8	67.0	60.6	65.0	59.9	63.8	67.1	62.5	64.9	61.4	63.6	7.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	85.3	64.8	61.4	81.7	63.6	60.3	78.4	62.6	29.1	67.1	28.2	64.9	27.4	63.9	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.7	25.9	22.8	DB	27.3	76.2	3.3	3.7	24.9	78.9	23.0	81.0	21.1	83.1	18.7	85.8	(4)
(5)			WB	26.1	66.3	3.3	2.5	23.7	68.0	21.8	69.5	19.9	70.9	17.5	72.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	45.6	45.6	46.6	49.1	52.9	56.7	59.5	59.6	57.6	53.8	49.1	46.7		
	DBStd	6.48	4.53	4.19	4.25	3.74	3.67	3.30	2.82	2.73	3.25	3.92	4.37	4.86		
	HDD50	661	143	131	118	58	12	0	0	0	0	13	67	119		
	HDD65	4780	600	544	569	478	377	251	174	171	223	348	478	567		
	CDD50	1363	8	6	14	30	101	201	293	296	228	130	39	17		
	CDD65	8	0	0	0	0	0	1	3	3	1	0	0	0		
	CDH74	24	0	0	0	0	2	8	6	8	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.5	12.5	12.0	11.0	10.1	9.9	9.2	8.6	8.8	9.3	10.5	11.1	12.6		
	PrecAvg	59.1	6.9	5.0	4.6	3.7	3.6	3.5	3.6	4.3	5.0	6.4	6.7	6.5		
(15) Precipitation	PrecMax	85.7	12.6	11.5	10.2	7.8	8.2	7.2	10.1	10.7	11.4	12.6	14.2	13.3		
	PrecMin	39.0	1.0	0.1	0.7	0.9	0.3	0.7	0.6	0.7	0.5	2.3	2.8	0.8		
	PrecStd	7.8	2.6	2.4	2.0	1.7	1.9	1.4	1.7	2.2	2.5	2.6	2.6	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	55.7	62.2	64.1	71.6	74.1	73.9	73.6	70.0	64.3	59.7	57.0		
		MCWB	53.3	52.1	51.2	54.2	60.3	62.7	64.4	64.7	60.8	58.9	56.5	54.7		
	2%	DB	53.9	53.8	55.6	59.5	64.9	68.6	68.8	67.9	66.2	61.9	57.8	55.6		
		MCWB	52.1	51.6	50.1	51.9	56.8	59.7	62.3	62.5	60.1	58.2	55.5	53.6		
	5%	DB	52.8	52.7	53.6	56.4	61.1	64.9	65.8	65.4	64.1	60.6	56.4	54.3		
		MCWB	51.1	50.7	50.4	51.1	55.1	58.6	61.0	61.0	59.7	57.6	54.2	52.5		
	10%	DB	51.9	51.7	52.5	54.6	58.5	62.3	64.0	63.9	62.7	59.4	55.2	53.1		
		MCWB	50.2	49.7	49.7	50.5	53.6	57.4	60.1	60.1	59.1	56.7	53.1	51.3		
	0.4%	WB	54.0	53.4	53.4	55.7	61.0	64.6	65.8	66.5	63.8	60.8	57.6	55.6		
		MCDB	54.5	53.9	54.9	60.4	68.8	72.6	71.7	71.6	66.3	62.5	58.5	56.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	52.8	52.2	52.4	53.8	58.0	61.1	63.5	63.9	62.3	59.5	56.4	54.2		
		MCDB	53.5	53.1	53.9	56.5	63.0	65.7	66.8	66.4	64.0	60.9	57.4	55.1		
	5%	WB	51.8	51.2	51.4	52.6	56.2	59.5	62.1	62.5	61.2	58.5	55.0	53.0		
		MCDB	52.6	52.3	53.0	55.1	59.6	63.0	64.7	64.3	63.0	59.9	56.0	54.0		
	10%	WB	50.6	50.0	50.4	51.4	54.7	58.3	60.9	61.2	60.0	57.3	53.6	51.8		
		MCDB	51.8	51.4	52.3	54.0	57.4	61.3	63.2	63.1	62.1	59.1	54.9	53.0		
	5% DB	MDBR	7.4	7.9	9.0	9.5	9.4	9.2	8.0	8.3	8.8	8.1	7.6	7.3		
		MCDBR	7.2	7.2	9.6	12.1	14.6	15.3	11.4	10.8	10.6	8.2	7.2	7.0		
		MCWBR	7.9	7.1	7.4	8.1	9.0	9.3	7.3	7.4	7.4	7.1	7.2	7.7		
		MCWBR	7.0	6.8	8.3	9.7	11.8	11.8	9.8	9.1	8.1	7.2	6.6	6.9		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.1	7.6	7.7	7.7	8.1	8.0	7.2	7.3	7.4	7.2	7.3	8.1		
	taub		0.361	0.364	0.384	0.388	0.400	0.378	0.384	0.369	0.368	0.359	0.351	0.354		
		taud	2.206	2.282	2.270	2.306	2.320	2.421	2.412	2.467	2.455	2.399	2.275	2.181		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		191	234	254	269	271	278	274	273	259	234	195	169		
		Edh at Noon	23	28	34	37	38	35	35	31	28	25	21	20		
	RadAvg		235	456	836	1353	1638	1769	1596	1367	988	571	290	176		
(45) All-Sky Solar Radiation	RadStd		22	35	73	90	88	153	133	92	78	47	22	18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page